Causal Discovery from Unseen Environments

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Abstract

Existing causal discovery methods using interventional data typically assume access to explicit environment labels and known intervention targets or, in lieu of environment metadata, rely on complicated and computationally intensive procedures—for instance, modeling environmental labels as latent variables or first inferring intervention targets in order to apply methods that require known labels. Surprisingly, we show that neither explicit environment labels nor auxiliary inference algorithms are strictly necessary: under random soft interventions, the classical ICA-LiNGAM algorithm can recover the causal graph by simply pooling interventional and observational data. We first demonstrate that, in linear Gaussian structural equation models where interventions independently perturb the noise distributions across nodes, pooling produces exactly independent non-Gaussian sources. This allows us to apply Independent Component Analysis (ICA) to recover the causal DAG without exact intervention target information and without environment labels. Nonetheless, the assumption of strictly-independent interventions can be overly restrictive; therefore, we relax this assumption and analyze the practically-motivated setting of single-node (atomic) soft interventions. While pooling data from atomic interventions induces the necessary non-Gaussianity, it also introduces source dependence, violating a core ICA assumption. However, theoretical analysis establishes that this dependence decays as $\Theta(1/d^2)$ with the number of variables d, whereas non-Gaussianity decays only as $\Theta(1/d)$. This yields a regime of mild misspecification at moderate d where sources are near-independent yet sufficiently non-Gaussian for ICA to succeed. In synthetic experiments, where data is pooled across many different environments, our label-free approach matches or exceeds label-aware methods. On the Sachs protein-signaling data, our approach orients edges more accurately than methods given known intervention targets (GIES, IGSP), and at least as accurately as methods given environment labels, although the latter recover the skeleton more accurately. Ultimately, our work challenges the assumed necessity of environment labels and complex inference mechanisms in interventional causal discovery and suggests a re-evaluation of the often-overlooked classical ICA-LiNGAM algorithm.

1 Introduction

Causal discovery is the task of recovering a directed acyclic graph (DAG) that encodes causal relationships between variables. In scientific applications, particularly in biology, causal discovery often uses interventional data in addition to observational data. For example, when inferring gene regulatory networks, interventions frequently involve CRISPR perturbations or gene knockdowns [Dixit et al., 2016, Replogle et al., 2022]. However, the relationship between an intended intervention and its actual effect is often *noisy*: off-target effects or failed knockdowns can render intervention labels unreliable. Explicitly verifying true intervention targets for each sample may require significant

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

experimental effort, whereas inference of intervention targets often requires large sample sizes and introduces additional methodological complexity [Varici et al., 2021]. Overall, the significant cost of obtaining accurate intervention labels raises the fundamental question of whether explicit intervention labels are *actually* necessary.

Many methods, such as GIES [Hauser and Bühlmann, 2012] and its descendants [Wang et al., 2017], assume that environment labels or intervention targets are known. Even when these labels are reliable, such methods compare data across environments using conditional-independence or invariance tests, which need many samples per environment. Their power therefore drops when environments are small or imbalanced, which becomes more likely as the number of environments grows [Gamella et al., 2025]. When labels are unavailable or unreliable, methods must instead recover the missing targets [Varici et al., 2021, Brouillard et al., 2020] or model interventions as latent variables, often through computationally expensive variational inference [Faria et al., 2022].

We propose a far simpler alternative based on the observation that a Gaussian variable subject to stochastic mean shifts becomes a (non-Gaussian) mixture of Gaussians. When interventions occur independently across nodes, the pooled data have mutually independent, non-Gaussian coordinates, meaning the data-generating process can be written as a Linear Non-Gaussian Acyclic Model (LiNGAM) [Shimizu et al., 2006]. Therefore, we can apply independent component analysis (ICA) [Comon, 1994] directly to the pooled data to recover the causal graph as in ICA-LiNGAM [Shimizu et al., 2006]. In this paper, we show that ICA-LiNGAM can perform label-free causal discovery on pooled interventional data. Our theoretical analysis quantifies the extent to which more realistic intervention scenarios deviate from LiNGAM's ideal assumptions, and demonstrates that ICA-LiNGAM recovers the underlying causal structure even under these mild misspecifications.

1.1 Problem Formulation

This section formalizes the idealized regime in which the pooling-data approach will theoretically work. In addition to the linear assumption on the structural equation model, LiNGAM relies on two critical assumptions on the additive noise: (1) the noise at each node must be independent across nodes, and (2) the noise must be non-Gaussian. Since these idealized conditions are unlikely to hold exactly in practice, we study a more realistic setting of single-node interventions in order to quantify the degree to which each assumption in the idealized setting is violated.

To analyze deviations from the independent-source assumption, we adopt the *atomic intervention* setting: one environment per vertex, each intervening on a single distinct node, with equal sample sizes across environments [Eberhardt et al., 2005]. Throughout, "atomic" refers only to the number of targets (one per environment); our interventions are *soft* mean shifts that leave the target's incoming dependencies intact, not hard (perfect) interventions that replace the target's structural equation. Pooling data across environments clearly violates assumption (1). Because each environment targets exactly one node, two nodes are never shifted in the same sample: knowing that node i was intervened on rules out an intervention on any node $j \neq i$. Designs that also intervene on several nodes at once move the pooled distribution toward independence, at least when they have no *excess co-occurrence*: they shift two nodes together no more often than independent targeting would. Random intervention failures also reduce the dependence by lowering the intervention rates. Among designs without excess co-occurrence, the atomic design is therefore the extreme case. Correlated off-target effects, which do have excess co-occurrence, can be more damaging (Appendix H), so the atomic design is an adverse reference case rather than a universal worst case.

To analyze violations of the non-Gaussianity assumption, we assume Gaussian additive noise at every node in the observational environment. This choice is without loss of generality for the purpose of our analysis: if the observational noise were already non-Gaussian, LiNGAM could be applied directly to observational data without any interventions, which is a strictly easier setting. By assuming Gaussianity in the observational environment, we ensure that any non-Gaussianity in the pooled data arises exclusively from the interventions themselves. This allows us to isolate the effect of intervention design on non-Gaussianity.

Inputs, outputs, and intuition. The method takes as *input* a single unlabeled data matrix: observational and interventional samples pooled together, with no record of which environment each sample came from or which node was targeted. Its *output* is a fully directed DAG with edge weights, obtained by running off-the-shelf ICA-LiNGAM on that matrix. For intuition, consider $x_1 \to x_2$

with Gaussian noise. From observational data alone the direction is not identifiable, since a bivariate Gaussian is symmetric in x_1 and x_2. If some samples receive a mean shift on x_1 and the labels are discarded, the pooled noise e_1 becomes a two-component Gaussian mixture, which is non-Gaussian, and the asymmetry ICA needs to orient the edge reappears.

1.2 Contributions

Under our model, the two LiNGAM assumptions are violated at different rates: total correlation (dependence) decays as $\Theta(1/d^2)$ (Theorem 3.1) while joint KL from Gaussianity decays as $\Theta(1/d)$ (Theorem 4.1). This creates a moderate-d regime where ICA-LiNGAM can succeed despite mild violations. We isolate each effect empirically: using Laplace noise (constant high non-Gaussianity) to measure dependence alone (Section 3.2), and Bernoulli interventions (independent sources) to measure non-Gaussianity alone (Section 4.2). Finally, we study the single-node design (both violations together) as a function of graph size, compare ICA-LiNGAM against label-aware baselines on synthetic graphs under independent-occurrence interventions (Section 5) and under the single-node design (Appendix F), and evaluate on the Sachs dataset. We also compare against methods that use environment labels but not intervention targets (JCI and Ψ-FCI), on synthetic data and on Sachs (Appendix G). We stress that our contribution is a characterization of *when* an existing algorithm suffices for label-free interventional discovery, not a new algorithm.

1.3 Related Works

Our work relates to algorithms for interventional causal discovery with known targets (GIES [Hauser and Bühlmann, 2012], IGSP [Wang et al., 2017], JCI [Mooij et al., 2020], k-RIC [Mazaheri et al., 2026b]), differentiable methods for unknown targets [Brouillard et al., 2020, Faria et al., 2022], identifiability of I-MECs [Yang et al., 2019], the LiNGAM framework [Shimizu et al., 2006], methods for unseen interventions and off-target effects [Squires et al., 2020, Kumar and Sinha, 2021, Jaber et al., 2020, Zhang et al., 2026, Kumar et al., 2024], mixtures of DAGs [Kumar et al., 2024, Saeed et al., 2020], and interventional discovery under latent confounding [Li et al., 2023, Magliacane et al., 2016, Hyttinen et al., 2014, Triantafillou and Tsamardinos, 2015]. A detailed discussion is deferred to Appendix A.

2 Preliminaries

2.1 Linear SCMs and ICA

We use the following conventions throughout. d denotes the number of observed variables (nodes in the causal graph). $\mathbf{x} = (x_1, \dots, x_d)^\top \in \mathbb{R}^d$ is the vector of observations and $\mathbf{e} = (e_1, \dots, e_d)^\top \in \mathbb{R}^d$ is the exogenous noise (sources). We consider a linear structural causal model

$$\mathbf{x} = A\mathbf{x} + \mathbf{e}, \qquad \text{equivalently} \qquad (I - A)\,\mathbf{x} = \mathbf{e}, \tag{1}$$

where $A \in \mathbb{R}^{d\times d}$ is the weighted adjacency matrix of a DAG with $A_{ji} \neq 0$ iff $x_i \to x_j$. Acyclicity guarantees that $(I - A)$ is invertible, so

$$\mathbf{x} = (I - A)^{-1}\mathbf{e}. \tag{2}$$

Equation (2) shows the data-generating process can be described by an independent component analysis (ICA) model [Comon, 1994, Hyvärinen and Oja, 2000]: an observation $\mathbf{x}$ is produced by linearly mixing latent sources $\mathbf{e}$ through the mixing matrix $(I - A)^{-1}$.

Comon [1994] showed that an ICA model is identifiable up to permutation and scaling of the sources provided that at most one source is Gaussian. Shimizu et al. [2006] observed that when the mixing matrix arises from a DAG as in Eq. (2), the acyclicity constraint resolves the permutation ambiguity: there is a unique row/column permutation under which $(I - A)$ is lower triangular with unit diagonal, recovering both a topological order and the edge weights. The resulting Linear Non-Gaussian Acyclic Model (LiNGAM) recovers the full DAG provided the noise components e_i are mutually independent and at most one is Gaussian.[1]

[1]The identifiability result relies on the Darmois–Skitovich theorem [Darmois, 1953, Skitovich, 1953], which states that if two linear combinations of independent random variables are independent, then every variable with a nonzero coefficient in both combinations is Gaussian.

2.2 Stochastic Mean-Shift Interventions

We model interventions as stochastic mean shifts on the noise. Consider environments $\mathcal{E} = \{0, 1, \ldots, d\}$, where 0 is the observational (un-intervened) setting and $\ell \in \{1, \ldots, d\}$ corresponds to an intervention on node ℓ. A sample drawn from environment ℓ replaces the noise e_ℓ at the intervened node with

$$\tilde{e}_\ell = e_\ell + \mu_\ell, \qquad \mu_\ell \sim \nu_\ell, \tag{3}$$

where ν_ℓ is the intervention's shift distribution, independent of $\{e_j\}_{j=1}^d$ and of the shifts in other environments. The pooled sample draws environments with weights $\{\pi_\ell\}_{\ell=0}^d$ summing to one.

For the single-node design, $\pi_\ell = 1/(d+1)$ uniformly across environments, and we absorb the observational environment into a uniform mixture of $d+1$ outcomes. In our main theoretical analysis we focus on this **atomic intervention design** of Section 1 (we use "single-node" and "atomic" interchangeably), an analytically tractable adverse reference case.

When samples are pooled, the marginal noise distribution at node i becomes a mixture

$$p(e_i) = (1 - \pi_i)\, \phi_i(e_i) + \pi_i\, (\phi_i * \nu_i)(e_i), \tag{4}$$

where ϕ_i is the un-intervened (Gaussian) noise density and $*$ denotes convolution (i.e., the density of $e_i + \mu_i$). Equation (4) is the central object of our analysis: a Gaussian mixed with a shifted version of itself, which is non-Gaussian whenever ν_i is non-degenerate. This is the mechanism by which interventions inject the non-Gaussianity that LiNGAM requires, even when each environment's noise is itself Gaussian. Under the atomic design, the events that node i is shifted and that node j is shifted are disjoint, so their indicators have covariance $-\pi_i \pi_j$, the minimum possible given these marginal rates (for $\pi_i + \pi_j \leqslant 1$); under independent co-occurrence it would be 0.

We also consider a simplified *Bernoulli intervention model* in which each node is independently intervened in each sample with probability $\pi \in (0, 1)$, yielding pooled coordinates that are exactly mutually independent and non-Gaussian. The two designs produce the same kind of non-Gaussian marginals but differ in their joint structure: under independent occurrence each disturbance depends only on its own noise and its own intervention indicator, so the product structure ICA requires is preserved exactly, whereas under the atomic design a single categorical variable selects at most one target, making the indicators mutually exclusive and the disturbances dependent. We use this as an idealized reference to isolate the effect of non-Gaussianity alone (Section 4.2); the formal model is deferred to Appendix B.

2.3 Diagnostics: Total Correlation and Excess Kurtosis

Our analysis tracks the two LiNGAM assumptions through two scalar diagnostics. The total correlation [Watanabe, 1960] of a random vector $\mathbf{e}$,

$$\mathrm{TC}(\mathbf{e}) = D_{\mathrm{KL}}\left(p(\mathbf{e}) \,\middle\|\, \prod_{i=1}^d p(e_i)\right) = \sum_{i=1}^d H(e_i) - H(\mathbf{e}), \tag{5}$$

is zero iff the components are mutually independent; we use it to quantify how far the pooled noise is from satisfying LiNGAM's independence assumption. The excess kurtosis of a scalar X with mean μ and variance σ^2,

$$\kappa_4(X) = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^4} - 3, \tag{6}$$

vanishes for any Gaussian and is a standard ICA contrast [Hyvärinen and Oja, 2000]; we use $|\kappa_4|$ to quantify how far the pooled noise is from being Gaussian.

As we will show in Sections 3 and 4, under the atomic design the total correlation decays as $\Theta(1/d^2)$ while the joint KL divergence from Gaussianity decays more slowly as $\Theta(1/d)$. This disparity creates a window at moderate d where ICA-LiNGAM can succeed even though both LiNGAM assumptions are violated. Comon's theorem gives *exact* identifiability whenever at most one source is Gaussian, but in finite samples small $|\kappa_4|$ degrades estimation; likewise, exact independence is rarely achieved, and small but nonzero TC similarly degrades ICA. Our analysis is therefore concerned with sample-regime behavior rather than asymptotic identifiability: when are both diagnostics favorable enough for ICA to recover the DAG accurately at realistic sample sizes? Note that under the atomic

design the dependence is a population-level property: at fixed d, more samples remove estimation error but not the dependence, so ICA-LiNGAM is not consistent; our rates identify a favorable regime at moderate d rather than asymptotic behavior in the sample size.

3 Deviations from Independence

LiNGAM identifiability requires the noise components $e_1, \ldots, e_d$ to be mutually independent. Pooling samples from single-node interventions (Section 2.2) breaks this assumption: only one node is shifted at a time, so the mixture components are perfectly anti-correlated in their assignment to environments. The questions for this section are how badly this dependence corrupts ICA-based recovery, and how it scales with the size of the graph.

We argue that dependence vanishes quickly as the graph grows. Section 3.1 proves that the aggregate total correlation of the pooled noise decays at rate $\Theta(1/d^2)$, driven by the cancellation of all sub-fourth-order moment contributions when the environment indicator is marginalized out. Section 3.2 verifies empirically that this is sufficient for ICA-LiNGAM to recover causal structure as accurately on pooled interventional data as on truly i.i.d. data.

3.1 Aggregate Total Correlation Decays as $\Theta(1/d^2)$

We work in a simplified model that is adverse for dependence among designs without excess co-occurrence. Let $\mathbf{u}_1, \ldots, \mathbf{u}_d$ denote the standard basis vectors in $\mathbb{R}^d$. Define $\Delta \sim \mathrm{Uniform}\{\mathbf{0}, \mathbf{u}_1, \ldots, \mathbf{u}_d\}$, independent of $Z \sim \mathcal{N}(0, I_d)$. Then $S = Z + \Delta$.

Here Δ either fires a unit shift on a single coordinate, chosen uniformly among the d axes, or contributes nothing (the $\Delta = 0$ outcome corresponds to the observational environment). This collapses the observational and interventional environments into a single uniform mixture of $d+1$ outcomes, which makes *every* environment correlated with every other. The argument extends with slight modifications to general $Z \sim \mathcal{N}(0, \mathrm{diag}(\sigma_1^2, \ldots, \sigma_d^2))$ and $\Delta \sim \mathrm{Uniform}\{\mathbf{0}, \delta_1 \mathbf{u}_1, \ldots, \delta_d \mathbf{u}_d\}$ with $\delta_i \neq 0 \in \mathbb{R}$.

Theorem 3.1 (Aggregate total correlation)**.** *For $S = Z + \Delta$ with $Z \sim \mathcal{N}(0, I_d)$ and $\Delta \sim \mathrm{Uniform}\{\mathbf{0}, \mathbf{u}_1, \ldots, \mathbf{u}_d\}$ independent,*

$$\mathrm{TC}(S) \;=\; D_{\mathrm{KL}}\left(P_S \,\middle\|\, \prod_{i=1}^{d} P_{S_i}\right) \;=\; \frac{(e-1)^2}{4d^2} + O(d^{-5/2}) \qquad (d \to \infty). \tag{7}$$

Proof sketch. The proof uses a change of measure, Taylor expansion of $g(u) = (1+u)\ln(1+u)$, and Rosenthal's inequality to control remainders; the leading $\Theta(1/d^2)$ term arises from fourth-order cancellations. See Appendix C. □

Interpretation. The $\Theta(1/d^2)$ rate is faster than what a naive analysis would suggest. Pairwise covariance between coordinates of S is $\Theta(1/d^2)$ (from a direct moment calculation), but pairwise covariance bounds aggregate dependence only at second order. The $\Theta(1/d^2)$ rate for total correlation is a stronger statement: it says that all sub-fourth-order multivariate cumulants of S are absorbed into the product of marginals, and the first non-vanishing residual is fourth-order. This is the substantive content of the cancellations in the proof. Theorem 3.1 implies that in the atomic design, dependence becomes negligible quickly as d grows so that for large d, it is unlikely to be a limiting factor in ICA.

3.2 LiNGAM Under Independence Violations

To link total correlation directly to LiNGAM's performance, we run ICA-LiNGAM [Shimizu et al., 2006] on pooled single-node intervention data over randomly sampled Erdős–Rényi DAGs. To isolate the effect of independence violation from non-Gaussianity, we use Laplace noise at every node so that all sources are highly non-Gaussian regardless of graph size; the only effect of varying d is then through the dependence induced by pooling environments. We sample 50 random DAGs at each graph size $d \in \{5, 6, 7, 8, 9, 10, 12, 15, 20, 30, 40, 50\}$ with edge probability $p = 2/d$, giving an average of two parents per node, and use $200d$ pooled samples (200 per environment). For each DAG we apply a random row permutation to disguise the topological order before fitting ICA-LiNGAM.

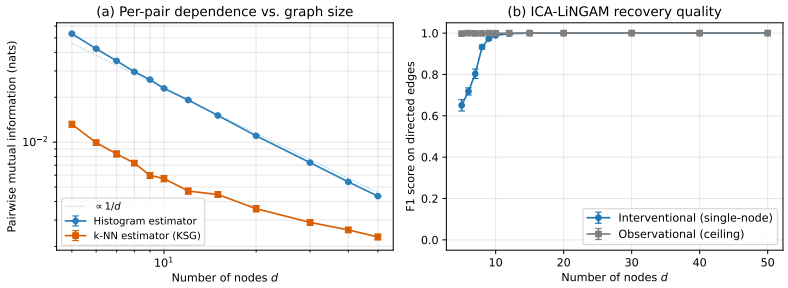


Figure 1: Isolating the effect of dependence on LiNGAM recovery. Highly non-Gaussian (Laplace) noise ensures the change in performance with d reflects only the dependence violation. Means and standard errors over 50 random Erdős–Rényi DAGs per graph size with $200d$ pooled samples. **(a)** Pairwise mutual information of the recovered noise estimated by histogram and k-NN estimators; both decay with d, with absolute values below 0.01 nats by $d \geqslant 10$ for the k-NN estimator. **(b)** F1 score of ICA-LiNGAM on pooled single-node intervention data approaches the observational ceiling by $d \approx 12$.

As a ceiling baseline, we also run ICA-LiNGAM on $200d$ i.i.d. observational samples from the same SCM (no interventions).

We measure (i) the average pairwise mutual information across coordinate pairs of the recovered noise, estimated via both a histogram plug-in and a k-nearest-neighbor estimator [Kraskov et al., 2004], and (ii) the F1 score on directed edges between recovered and true DAGs. Average excess kurtosis was monitored as a sanity check on the experimental design and remained high across all d, confirming that non-Gaussianity is held approximately constant.

Results. Figure 1 summarizes the results. Two observations support the central claim of this section.

First, pairwise dependence in the pooled noise is small at all d and decays smoothly with d (Fig. 1a). Both estimators report pairwise mutual information well below 0.05 nats across the entire range, with the k-NN estimator below 0.01 nats by $d \geqslant 10$. Since pairwise mutual information is bounded above by total correlation, these values are consistent with Theorem 3.1, although neither estimator resolves its $\Theta(1/d^2)$ rate cleanly at our sample sizes.

Second, recovery quality on interventional data converges to the observational ceiling by $d \approx 12$ (Fig. 1b). ICA-LiNGAM run on i.i.d. observational data gives F1 close to 1.0 across all d because we are in the regime in which its assumptions hold. Recovery on pooled single-node intervention data has a visible gap at $d = 5$ (F1 $= 0.65$) that closes monotonically: $0.72, 0.80, 0.93, 0.97, 0.99$ at $d = 6, 7, 8, 9, 10$, reaching the ceiling set by the fully valid regime by $d = 12$ and remaining there for all larger graphs.

4 Deviations from Non-Gaussianity

LiNGAM identifiability requires the noise components $e_1, \ldots, e_d$ to be non-Gaussian. The pooled noise satisfies this because each marginal is a Gaussian mixture that is non-Gaussian whenever the intervention is non-degenerate. In this section, we characterize the rate at which the "amount of non-Gaussianity" decays as the graph grows: aggregate non-Gaussianity (joint KL from the Gaussian baseline) shrinks as $\Theta(1/d)$, which is an order slower than the dependence rate $\Theta(1/d^2)$ from Section 3.1. We then verify empirically that this trade-off opens a window at moderate d where ICA-LiNGAM recovers structure despite both violations.

4.1 Aggregate Non-Gaussianity Decays as $\Theta(1/d)$

The same change-of-measure technique used in Section 3.1 bounds the joint KL divergence between S and a standard Gaussian baseline.

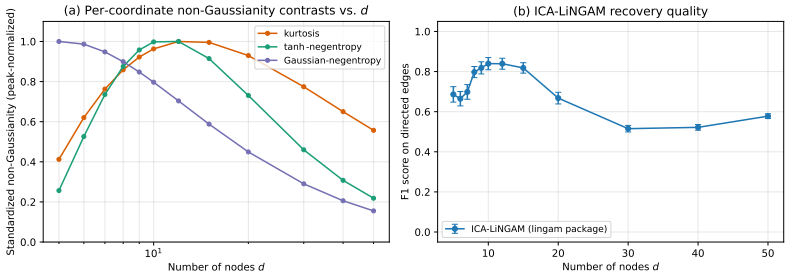


Figure 2: Isolating the effect of non-Gaussianity on LiNGAM recovery. Sources are Gaussian and interventions occur independently with probability $1/d$ per node. Means and standard errors over 50 random ER DAGs per d with $200d$ pooled samples. **(a)** Three standardized non-Gaussianity contrasts in closed form for the pooled marginals, peak-normalized for visual comparison. **(b)** F1 score of ICA-LiNGAM (FastICA with `logcosh`).

Theorem 4.1 (Joint KL from the Gaussian baseline)**.** *Under the assumptions of Theorem 3.1,*

$$D_{\mathrm{KL}}(S \,\|\, Z) = \frac{e-1}{2d} + O(d^{-3/2}) \qquad (d \to \infty).$$

Proof idea. By the same change of measure as in Theorem 3.1, $D_{\mathrm{KL}}(S \,\|\, Z) = \mathbb{E}_Z[g(\epsilon Y)]$. Taylor-expanding g to second order, the first-order term vanishes by $\mathbb{E}[Y] = 0$ and the second-order term gives $\frac{1}{2}\mathbb{E}[(\epsilon Y)^2] = \frac{e-1}{2d} + O(d^{-2})$. Tail probabilities and remainders are controlled at $O(d^{-3/2})$ via Rosenthal's inequality. See Appendix D. □

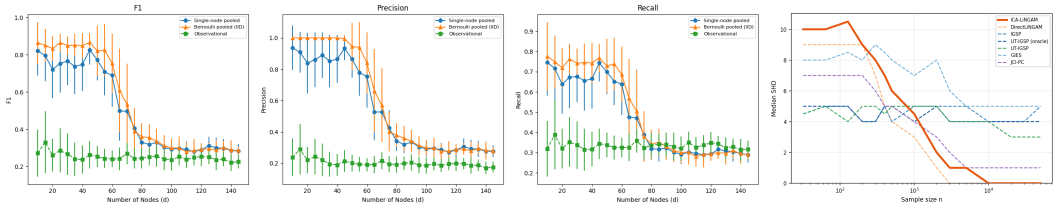
Comparison with $\mathrm{TC}(S)$. The joint KL decays as $\Theta(1/d)$, an order slower than the total correlation $\Theta(1/d^2)$ from Theorem 3.1. The pooled distribution thus departs from joint Gaussianity at rate $1/d$ but departs from being a product of marginals at rate $1/d^2$: the marginals absorb the leading $1/d$ non-Gaussianity, leaving only a $1/d^2$ residual of cross-coordinate structure. This trade-off is exploited for the rest of the paper: non-Gaussianity (good for ICA) decays as $1/d$ while dependence (bad for ICA) decays as $1/d^2$.

4.2 LiNGAM Under Non-Gaussianity Violation

Although Theorem 4.1 shows that aggregate non-Gaussianity decays monotonically as $\Theta(1/d)$, the contrast functions used in practice by ICA algorithms are *standardized*: they are computed after rescaling each variable to zero mean and unit variance, so they depend only on the shape of its distribution, and every Gaussian, whatever its mean and variance, has zero contrast. This standardization causes a contrast to vanish at *both* mixture limits—as $\pi \to 0$ the un-intervened component dominates and the marginal becomes a single Gaussian, and as $\pi \to 1$ the intervened component dominates and the marginal becomes a different (but still single) Gaussian. The result is a non-monotonic curve in d (since $\pi = 1/d$), with a maximum at some intermediate d^*.

Setup. To isolate the non-Gaussianity violation from dependence, we depart from the single-node design and use *independent-occurrence interventions*: each node fires its intervention independently in each sample with probability $\pi = 1/d$, ensuring the pooled coordinates are mutually independent. We sample 50 random Erdős–Rényi DAGs at each $d \in \{5, 6, 7, 8, 9, 10, 12, 15, 20, 30, 40, 50\}$ with edge probability $2/d$, and use $200d$ pooled samples per DAG. We compute three standardized non-Gaussianity contrasts—excess kurtosis, the $\tanh$-based negentropy approximation used by FastICA's default `logcosh` option, and a Gaussian-derivative-based negentropy approximation—in closed form for the pooled marginals, and measure F1 score on directed edges of recovered DAGs.

Results. Figure 2(a) shows that all three contrasts are non-monotonic with peaks at intermediate d, with peak locations differing across contrasts ($d \approx 13$ for kurtosis, $d \approx 10$ for $\tanh$-negentropy, $d \approx 5$ for Gaussian-negentropy at $\mu = 2, \tau = 0.5, \sigma = 1$); the aggregate $\Theta(1/d)$ rate of Theorem 4.1 is recovered at large d for all three. Panel (b) shows that ICA-LiNGAM's recovery quality tracks the curve of its underlying contrast: F1 stays high where $\tanh$-negentropy is high and falls steeply



(a) Recovery vs. graph size d

(b) SHD vs. sample size at $d = 10$

Figure 3: Synthetic evaluation. **(a)** F1, precision, and recall versus d for three pooling regimes: single-node interventions (dependent non-Gaussian), Bernoulli independent-occurrence interventions with probability $1/d$ (independent non-Gaussian), and observational data from the un-intervened linear Gaussian SCM. The dependent curves converge to the independent curves as d grows; both eventually degrade as non-Gaussianity collapses. **(b)** ICA-LiNGAM on pooled data from independent interventions outperforms label-aware baselines (GIES, IGSP, UT-IGSP) at smaller sample sizes and remains consistent as samples grow. An additional ablation on the kurtosis–recovery relationship under the single-node design appears in Appendix F.

once it collapses. The peak of non-Gaussianity defines a *sweet spot* for ICA-based recovery; outside this regime, recovery degrades rapidly. It degrades fastest on the large-d side, where non-Gaussianity becomes too weak even though dependence has been removed by design.

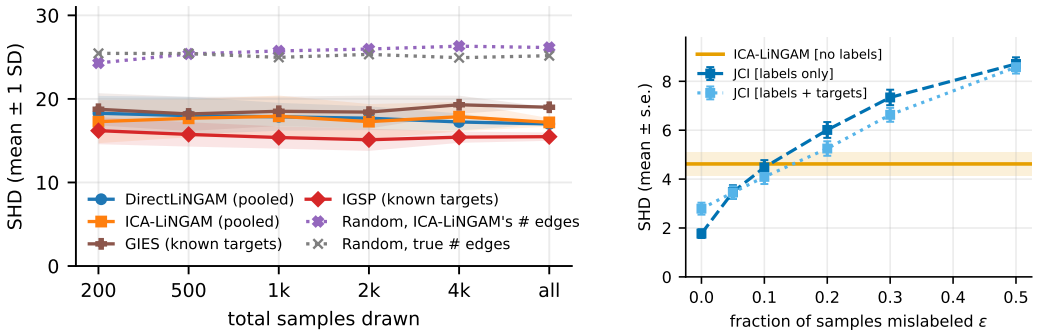
5 Synthetic and Real-Data Experiments

Sections 3.1 and 4.1 bound the two violations separately—total correlation as $\Theta(1/d^2)$ and joint KL as $\Theta(1/d)$—and Sections 3.2 and 4.2 verify each effect in isolation. The more realistic single-node design exhibits both violations, trading one off against the other as the graph grows. In this section, we study how that trade-off affects ICA-LiNGAM's performance and compare ICA-LiNGAM with label-aware baselines on synthetic and real-world data.

5.1 Synthetic Experiments

Setup. All synthetic graphs are random Erdős–Rényi DAGs with edge probability $2/d$ and edge weights drawn from $[-2, -0.5] \cup [0.5, 2]$. For the graph-size sweep (panel a), we vary d and run ICA-LiNGAM on three pooling regimes: single-node interventions (dependent, non-Gaussian), Bernoulli independent-occurrence interventions with probability $1/d$ (independent, non-Gaussian), and observational data from the un-intervened linear Gaussian SCM (independent, Gaussian). The Bernoulli design serves as a matched control: it has the same per-node intervention rate as the single-node design but independent sources, so the gap between the two curves isolates the effect of dependence. The observational case is used as a reference because ICA-LiNGAM is not expected to work there. Interventions shift the mean of target i by $2\,\widehat{\mathrm{Var}}(X_i)$, where $\widehat{\mathrm{Var}}(X_i)$ is the empirical observational variance of X_i, and we use $\min(10^4 d,\ 50{,}000)$ pooled samples and 20 DAGs per graph size. For the sample-size sweep (panel b), we fix $d = 10$ and generate independent interventions per node with probability $1/2$ that shift the mean by $\widehat{\mathrm{Var}}(X_i)$. We vary the total sample size, comparing ICA-LiNGAM against label-aware baselines (GIES [Hauser and Bühlmann, 2012], IGSP [Wang et al., 2017], UT-IGSP [Squires et al., 2020]), and average over 100 DAGs at each sample size. ICA-LiNGAM operates on pooled data without environment labels; GIES and IGSP receive environment labels and known intervention targets, and UT-IGSP receives environment labels only.

Results. Figure 3(a) shows that the dependent and independent interventional curves converge as d grows, consistent with the $\Theta(1/d^2)$ decay of total correlation, and both eventually degrade for large d as non-Gaussianity collapses at rate $\Theta(1/d)$. The Gaussian observational baseline tracks neither, confirming that interventions, not graph size, are providing the non-Gaussianity that ICA exploits. Figure 3(b) shows that ICA-LiNGAM matches or outperforms label-aware baselines at smaller sample sizes and remains consistent as samples grow. The label-aware baselines plateau



(a) Sachs: SHD vs. sample size

(b) Synthetic: label noise, $n = 10^4$

Figure 4: **(a)** Mean SHD on the Sachs protein signaling network as a function of total sample size. ICA-LiNGAM on the pooled 9-condition data is competitive with label-aware baselines (GIES, IGSP) on the 5-condition subset of Wang et al. [2017], despite using no environment labels. Random-guess baselines (matched to ICA-LiNGAM's recovered edge count and to the true edge count) shown for reference. Means $\pm$ 1 SD over 100 random subsamples drawn without replacement. **(b)** Single-node design, 10^4 samples: SHD as a fraction ε of samples is mislabeled (mean $\pm$ s.e., 100 DAGs; details in Appendix G.3).

because their per-environment subsamples stay small even as the total pool grows. ICA-LiNGAM derives significantly more benefit from the additional samples.

Labels-only baselines. Under the single-node design (Appendix G.1), ICA-LiNGAM has the lowest SHD at small n, but labels-only JCI [Mooij et al., 2020] overtakes it near 10^3 interventional samples, since the atomic design's dependence does not vanish with n (Section 2.3). That advantage depends on the labels being nearly exact: assigning 10–20% of samples to the wrong environment erases it (Figure 4b), whereas errors in the listed targets matter much less (Appendix G.3).

5.2 Real Data: Sachs Protein Signaling

We evaluate on the Sachs et al. protein signaling dataset [Sachs et al., 2005], a benchmark for causal discovery from interventional flow cytometry measurements with 11 phosphorylated proteins and 9 perturbation conditions. ICA-LiNGAM operates on the pooled data without environment labels while baselines use label information. Figure 4 sweeps total sample size, drawing samples asymmetrically across conditions. ICA-LiNGAM uses the full 9-condition pool of 6456 samples; GIES and IGSP use the 4275-sample subset of 5 interventional conditions plus the observational one (the "5-condition subset") following Wang et al. [2017] because the other 3 conditions (a co-stimulus and two pathway activators) have broad effects that do not correspond to a single known target. We find that ICA-LiNGAM's mean SHD against the consensus ground truth network is comparable to label-aware baselines and stable across sample sizes, despite operating without environment information. Labels-only JCI and Ψ-FCI [Jaber et al., 2020] obtain lower SHD than ICA-LiNGAM, mainly through a better skeleton; ICA-LiNGAM's directed-edge F1 (0.35–0.40) exceeds that of GIES, IGSP, and Ψ-FCI (at most 0.24) and is matched only by JCI at $n = 2000$–4000 (Appendix G.2).

6 Discussion and Limitations

Our results establish that the classical ICA-LiNGAM algorithm, applied to pooled interventional data without environment labels, recovers causal structure in a moderate-d regime where intervention-induced non-Gaussianity is sufficient and intervention-induced dependence is small. On synthetic data, this label-free approach matches or exceeds label-aware baselines (GIES, IGSP, UT-IGSP); on the Sachs benchmark its SHD is higher than IGSP's, but it orients edges more accurately than both GIES and IGSP. Labels-only methods (JCI, Ψ-FCI) are stronger competitors at large n and on the Sachs skeleton (Appendix G), so dropping labels pays off mainly when samples are scarce or labels are unavailable or unreliable. We do not claim state-of-the-art performance on real biological data,

but we view the synthetic results as evidence that the approach has substantive promise once the regime is well-matched.

Several extensions of our setup are natural targets for future work. Multi-node and stochastically-targeted interventions are natural examples: by shifting several nodes in the same sample at least some of the time, they weaken the mutual exclusivity that drives dependence in the single-node design, even with moderate excess co-occurrence. They also raise the per-node intervention probability. Both effects extend the sweet spot upward in d and may shift the peak to regimes where ICA contrasts are stronger. Characterizing how the sweet spot depends jointly on the intervention design, off-target rate, and contrast choice is a concrete open question. A second direction we find conceptually interesting: *discarding* environment labels can improve recovery in some regimes, even though labels add information. This observation may be useful for causal representation learning, where multi-environment data is a standard ingredient and the question of when to condition on environment versus when to pool is rarely addressed. We hope our analysis encourages closer attention to this trade-off in that setting.

Finally, existing results [Jaber et al., 2020, Squires et al., 2020] rely on interventional faithfulness to establish the identifiability of the equivalence class of causal graphs from unknown interventions. Our approach instead assumes a linear functional form and relies on the parametric form of the noise to allow for identifiability in the independent noise case without needing I-faithfulness. Mazaheri et al. [2026b] take a different route, using hard interventions as the primary source of structural information to relax faithfulness. A full characterization of the relative identifiability boundaries is left for future work.

Limitations We assume a linear structural equation model with Gaussian noise in the observational environment and mean-shift interventions. We also assume *causal sufficiency*: there are no hidden confounders. This is a substantive assumption separate from linearity and Gaussianity, since latent confounding induces dependence among the observed disturbances and changes the source model; methods that allow latent confounders in the interventional setting include ACI [Magliacane et al., 2016], HEJ [Hyttinen et al., 2014], COmbINE [Triantafillou and Tsamardinos, 2015], S-FCI [Li et al., 2023], and Ψ-FCI [Jaber et al., 2020]. On Sachs, Ψ-FCI returns mostly bidirected edges (Appendix G.2), suggesting this assumption is violated there. Our analysis further requires a structural matrix A shared across environments. When unlabeled environments instead change the mechanisms on the same graph, the environment acts as a latent confounder of every variable and pooling makes discovery harder rather than easier [Mazaheri et al., 2026a]. Hard (perfect) interventions also violate this, since they remove the target's incoming edges, so pooling yields a mixture of distinct LiNGAM models rather than a single mixing matrix; recovering the graph from such data is outside the scope of ICA-LiNGAM and is better addressed by mixture-of-DAGs methods [Kumar et al., 2024, Saeed et al., 2020]. If the underlying system is non-linear or the observational noise is already non-Gaussian, the pooling approach may still work (non-Gaussianity would only help), but our theoretical rates no longer apply. Extending the analysis to non-linear models or non-Gaussian observational noise is an important direction for future work. Also, theoretical rates are derived for the atomic intervention design. Realistic experiments may have uneven sample sizes, different shift magnitudes across nodes, or simultaneous multi-node interventions. While we argue that the atomic setting provides a worst-case bound on dependence for designs without excess co-occurrence, we have not characterized the dependence rate for arbitrary off-target or multi-node effects, and correlated off-target effects substantially degrade recovery (Appendix H). Practitioners should check that their intervention design has no strong excess co-occurrence, such as correlated off-target effects, or use diagnostics (e.g., estimated excess kurtosis) to assess whether non-Gaussianity is sufficiently strong.

Acknowledgments and Disclosure of Funding

B.M. and S.X. are partially supported by the Advanced Research Concepts (ARC) COMPASS program, sponsored by the Defense Advanced Research Projects Agency (DARPA) under agreement number HR00110112530212. The authors declare no competing interests. **Use of AI tools.** Part of the experiment code was written by the authors and checked by Claude Opus 5.5 (Anthropic); the remaining experiments were implemented by Claude Opus 5.5, which also modified parts of the author-written code. The text was written by the authors and polished with AI assistance.

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A Related Works

Algorithms for interventional causal discovery GIES [Hauser and Bühlmann, 2012] extends greedy equivalence search to interventional data under hard interventions (Wang et al. [2017] showed that it is not guaranteed to be consistent). IGSP [Wang et al., 2017] is a nonparametric method that uses invariance-based tests across environments. Joint causal inference (JCI) [Mooij et al., 2020] encodes environments as auxiliary variables in a single graph.
Brouillard et al. [2020] develop a differentiable method of causal discovery from interventional and observational data. Building on continuous constrained optimization methods for finding a DAG, Faria et al. [2022] propose a method for causal discovery under fully latent interventions, through a neural-based variational approach that infers the correspondences between data samples and interventions.

Identifiability results in the setting of causal discovery from interventional and observational data The line of work on I-MECs [Yang et al., 2019] characterizes the identifiability gains from interventional data when targets are known.

ICA and LiNGAM The LiNGAM framework [Shimizu et al., 2006] leverages the Darmois–Skitovich theorem [Darmois, 1953, Skitovich, 1953] to identify linear SEMs with non-Gaussian noise. ICA-LiNGAM uses ICA to identify a topological ordering.

Existing work on modeling unseen interventions and off-target effects UT-IGSP [Squires et al., 2020] extends IGSP to unknown targets and characterizes the equivalence class of DAGs that remains identifiable when the targets are unknown. Kumar and Sinha [2021] develop identifiability theory and algorithms for recovering intervention targets in the case of unseen interventions. Jaber et al. [2020] formulate a graphical characterization to test whether two pairs of causal graphs and their

corresponding interventional target sets are in the same ψ-Markov equivalence class, i.e., they are indistinguishable with respect to the available datasets. Zhang et al. [2026] uses variational methods with latent intervention variables at substantial computational cost. Kumar et al. [2024] recovers causal structure by disentangling environments via a Gaussian mixture model, then learning labels.

Mixtures of DAGs When interventions change the causal mechanisms themselves (e.g., hard interventions that remove incoming edges), pooled data is a mixture of distinct DAGs rather than a single DAG with a mixed noise distribution. This setting is studied by Saeed et al. [2020] and Kumar et al. [2024]; our analysis instead assumes a structural matrix shared across environments.

Interventional discovery with latent confounders Several methods combine observational and interventional data while allowing hidden confounders, including ACI [Magliacane et al., 2016], HEJ [Hyttinen et al., 2014], COmbINE [Triantafillou and Tsamardinos, 2015], S-FCI [Li et al., 2023], and Ψ-FCI [Jaber et al., 2020]. FCI-family methods return a MAG or PAG; our method instead assumes causal sufficiency, which is what permits committing to a fully directed DAG. A related setting arises when unlabeled populations share a single DAG but differ in their causal mechanisms: the population then acts as a global latent confounder, which Mazaheri et al. [2026a] show can still be identified when the number of populations is small relative to the size and sparsity of the graph.

B Bernoulli Intervention Model

We will compare to a simplified data-generating process in which the pooled coordinates are *exactly* mutually independent. The observational data-generating process is a linear Gaussian SCM, and interventions are soft mean shifts applied to the exogenous noise, occurring independently at each node with probability $\pi \in (0, 1)$.

Concretely, we observe

$$\mathbf{x} = A\mathbf{x} + \mathbf{e}, \qquad e_i = \delta_i B_i + Z_i, \tag{8}$$

where $\delta_i \in \mathbb{R} \setminus \{0\}$ is the mean-shift magnitude for node i, $B_i \sim \text{Bernoulli}(\pi)$, and $Z_i \sim \mathcal{N}(0, \sigma_i^2)$. The $\{B_i\}$ and $\{Z_i\}$ are independent across coordinates and independent of each other.

The marginal distribution of e_i is a two-component Gaussian mixture,

$$e_i \;\sim\; (1-\pi)\,\mathcal{N}(0, \sigma_i^2) \;+\; \pi\,\mathcal{N}(\delta_i, \sigma_i^2), \tag{9}$$

and the joint distribution of $\mathbf{e} = (e_1, \ldots, e_d)$ is a 2^d-component Gaussian mixture indexed by $b \in \{0,1\}^d$:

$$\mathbf{e} \;\sim\; \sum_{b \in \{0,1\}^d} \left[\prod_{i=1}^{d} \pi^{b_i} (1-\pi)^{1-b_i} \right] \mathcal{N}(\delta \odot b,\, \Sigma), \tag{10}$$

where $\Sigma = \text{diag}(\sigma_1^2, \ldots, \sigma_d^2)$ and $\delta \odot b$ denotes the elementwise product. The coordinates of $\mathbf{e}$ are mutually independent, since each e_i depends only on (B_i, Z_i) and these pairs are independent across i.

Identifiability. The marginals of $\mathbf{e}$ are non-Gaussian (each is a two-component Gaussian mixture with $\delta_i \neq 0$) and mutually independent. By Darmois–Skitovich [Darmois, 1953], the ICA model $\mathbf{x} = (I - A)^{-1}\mathbf{e}$ is identifiable up to permutation and scaling. Combined with the acyclicity of A, the LiNGAM identification [Shimizu et al., 2006] recovers A exactly from the pooled data without environment labels.

C Proof of Theorem 3.1

We restate the theorem for convenience. Let $Z \sim \mathcal{N}(0, I_d)$ and $\Delta \sim \text{Uniform}\{\mathbf{0}, \mathbf{u}_1, \ldots, \mathbf{u}_d\}$ be independent, and set $S = Z + \Delta$. Then

$$\text{TC}(S) = \frac{(e-1)^2}{4d^2} + O(d^{-5/2}) \qquad (d \to \infty).$$

C.1 Setup and Change of Measure

Let ϕ denote the standard Gaussian density on $\mathbb{R}^d$, so that $Z \sim \phi$. From the definition of S, the densities are

$$P_S(x) = \frac{1}{d+1}\Big(\phi(x) + \sum_{i=1}^{d} \phi(x - \mathbf{u}_i)\Big), \qquad P_{S_i}(x_i) = \frac{d}{d+1}\phi_1(x_i) + \frac{1}{d+1}\phi_1(x_i - 1), \tag{11}$$

where ϕ_1 is the one-dimensional standard Gaussian density. Define the likelihood ratios

$$L(x) = \frac{P_S(x)}{\phi(x)}, \qquad L_i(x_i) = \frac{P_{S_i}(x_i)}{\phi_1(x_i)}.$$

Using $\phi(x - \mathbf{u}_i)/\phi(x) = \exp(x_i - 1/2)$,

$$L(x) = \frac{1}{d+1}\Big(1 + \sum_{i=1}^{d} e^{x_i - 1/2}\Big), \qquad L_i(x_i) = \frac{d + e^{x_i - 1/2}}{d+1}. \tag{12}$$

Set $\epsilon = 1/(d+1)$ and define the centered log-normal variables $U_i = e^{Z_i - 1/2} - 1$, where $Z_i \sim \mathcal{N}(0,1)$ are i.i.d. Then $L(Z) = 1 + \epsilon Y$ with $Y = \sum_{i=1}^{d} U_i$, and $L_i(Z_i) = 1 + \epsilon U_i$.

C.2 Properties of the U_i

The variables U_i are i.i.d. shifted log-normal, with the following properties:

- $\mathbb{E}[U_i] = 0$, since $\mathbb{E}[e^{Z_i - 1/2}] = e^{1/2 - 1/2} = 1$.
- $\mathbb{E}[U_i^2] = \mathrm{Var}(e^{Z_i - 1/2}) = e^{2-1} - 1 = e - 1$.
- $\mathbb{E}[|U_i|^p] < \infty$ for all $p \geqslant 1$, since U_i has all moments of the log-normal distribution.

C.3 Exact Identity for Total Correlation

Because $L = dP_S/d\phi$ is the Radon–Nikodym derivative, $\mathbb{E}_S[f(S)] = \mathbb{E}_Z[L(Z)f(Z)]$ for any measurable f. Applying this to $\mathrm{TC}(S) = \mathbb{E}_S[\ln L(S) - \sum_i \ln L_i(S_i)]$,

$$\mathrm{TC}(S) = \mathbb{E}_Z\Big[L(Z)\Big(\ln L(Z) - \sum_{i=1}^{d} \ln L_i(Z_i)\Big)\Big] = \mathbb{E}_Z[L(Z)\ln L(Z)] - \sum_{i=1}^{d} \mathbb{E}_Z[L(Z)\ln L_i(Z_i)]. \tag{13}$$

Define $g(u) = (1+u)\ln(1+u)$ for $u > -1$. The first term in Eq. (13) equals $\mathbb{E}_Z[g(\epsilon Y)]$. For the marginal terms, write $1 + \epsilon Y = 1 + \epsilon U_i + \epsilon \sum_{j \neq i} U_j$, so

$$L(Z)\ln L_i(Z_i) = (1+\epsilon U_i)\ln(1+\epsilon U_i) + \epsilon\Big(\sum_{j\neq i} U_j\Big)\ln(1+\epsilon U_i) = g(\epsilon U_i) + \epsilon\Big(\sum_{j\neq i} U_j\Big)\ln(1+\epsilon U_i).$$

By independence and $\mathbb{E}[U_j] = 0$, the second term vanishes in expectation. We obtain the exact identity

$$\mathrm{TC}(S) = \mathbb{E}_Z[g(\epsilon Y)] - \sum_{i=1}^{d} \mathbb{E}_Z[g(\epsilon U_i)]. \tag{14}$$

C.4 Moment Bounds

By Rosenthal's inequality [Rosenthal, 1970], for any $p \geqslant 2$,

$$\mathbb{E}[|Y|^p] \leqslant C_p \max\Big(d\,\mathbb{E}[|U_1|^p],\ (d\,\mathbb{E}[U_1^2])^{p/2}\Big) \leqslant C_p' d^{p/2},$$

since the variance term dominates for large d. Set $C_p' = C_p(e-1)^{p/2}$.

Define the typical set $A_d = \{|\epsilon Y| < 1/2\}$. Markov's inequality with $p = 10$ gives

$$\mathbb{P}(A_d^c) = \mathbb{P}(|\epsilon Y| \geqslant 1/2) \leqslant 2^{10}\epsilon^{10}\mathbb{E}[|Y|^{10}] = O(d^{-5}). \tag{15}$$

C.5 Atypical-Set Contributions

We show that the contributions from A_d^c to Eq. (14) are negligible.

Joint term. For all $y > -1$, one verifies that $g(y)^2 \leqslant 2(y^2 + y^4) + 2$. Hence

$$\mathbb{E}[g(\epsilon Y)^2] \leqslant 2\mathbb{E}[(\epsilon Y)^2] + 2\mathbb{E}[(\epsilon Y)^4] + 2.$$

Using $\mathbb{E}[(\epsilon Y)^2] = \epsilon^2 d(e-1) = O(d^{-1})$ and $\mathbb{E}[(\epsilon Y)^4] = O(d^{-2})$ from Rosenthal, $\mathbb{E}[g(\epsilon Y)^2] = O(1)$. Cauchy–Schwarz then gives

$$\left|\mathbb{E}[g(\epsilon Y)\mathbf{1}_{A_d^c}]\right| \leqslant \sqrt{\mathbb{E}[g(\epsilon Y)^2]} \cdot \sqrt{\mathbb{P}(A_d^c)} = O(1) \cdot O(d^{-5/2}) = O(d^{-5/2}). \tag{16}$$

Marginal terms. For the marginal terms, we need a sharper L^2 estimate. On $|y| \leqslant 1$, one verifies $|g(y)| \leqslant 2|y|$ (a direct calculation using $g(0) = 0$ and the bound $|g'(\xi)| \leqslant 1 + |\ln 2| < 2$ for $\xi \in (-1/2, 1)$). For $|y| > 1$, we use the polynomial bound $g(y)^2 \leqslant 2(y^2 + y^4) + 2$. Splitting the expectation:

$$\begin{aligned}\mathbb{E}[g(\epsilon U_i)^2] &\leqslant 4\epsilon^2 \mathbb{E}[U_i^2 \mathbf{1}_{|\epsilon U_i| \leqslant 1}] + \mathbb{E}[(2(\epsilon U_i)^2 + 2(\epsilon U_i)^4 + 2)\mathbf{1}_{|\epsilon U_i| > 1}] \\ &\leqslant 4\epsilon^2(e-1) + O(\mathbb{P}(|\epsilon U_i| > 1)).\end{aligned}$$

Since U_i has all moments finite, $\mathbb{P}(|\epsilon U_i| > 1) = \mathbb{P}(|U_i| > d+1)$ decays super-polynomially in d. Therefore $\mathbb{E}[g(\epsilon U_i)^2] = O(d^{-2})$. Cauchy–Schwarz gives

$$\left|\mathbb{E}[g(\epsilon U_i)\mathbf{1}_{A_d^c}]\right| \leqslant \sqrt{\mathbb{E}[g(\epsilon U_i)^2]} \cdot \sqrt{\mathbb{P}(A_d^c)} = O(d^{-1}) \cdot O(d^{-5/2}) = O(d^{-7/2}).$$

Summing over i,

$$\left|\sum_{i=1}^{d} \mathbb{E}[g(\epsilon U_i)\mathbf{1}_{A_d^c}]\right| = d \cdot O(d^{-7/2}) = O(d^{-5/2}). \tag{17}$$

C.6 Taylor Expansion on the Typical Set

On A_d, $|\epsilon Y| < 1/2$, so we may Taylor-expand g around zero. Computing derivatives:

$$g(0) = 0, \quad g'(0) = 1, \quad g''(0) = 1, \quad g'''(0) = -1, \quad g^{(4)}(0) = 2,$$

and $|g^{(5)}(\xi)| \leqslant 96$ for $|\xi| \leqslant 1/2$. Lagrange's form of the remainder gives

$$g(u) = u + \tfrac{1}{2}u^2 - \tfrac{1}{6}u^3 + \tfrac{1}{12}u^4 + R_4(u), \qquad |R_4(u)| \leqslant \tfrac{96}{120}|u|^5 = \tfrac{4}{5}|u|^5 \tag{18}$$

for $|u| \leqslant 1/2$. Define $P_4(u) = u + \frac{1}{2}u^2 - \frac{1}{6}u^3 + \frac{1}{12}u^4$.

Cancellation of low-order moments. For $k = 1, 2, 3$, independence and zero means give $\mathbb{E}[Y^k] = \sum_i \mathbb{E}[U_i^k]$:

- $\mathbb{E}[Y] = \sum_i \mathbb{E}[U_i] = 0 = \sum_i \mathbb{E}[U_i]$.
- $\mathbb{E}[Y^2] = \sum_i \mathbb{E}[U_i^2]$ since cross-terms $\mathbb{E}[U_iU_j] = \mathbb{E}[U_i]\mathbb{E}[U_j] = 0$.
- $\mathbb{E}[Y^3] = \sum_i \mathbb{E}[U_i^3]$ since cross-terms involve at least one factor with mean zero.

At fourth order,

$$\mathbb{E}[Y^4] = \sum_i \mathbb{E}[U_i^4] + 6\sum_{i<j} \mathbb{E}[U_i^2]\mathbb{E}[U_j^2] = d\,\mathbb{E}[U_1^4] + 3d(d-1)(e-1)^2.$$

Hence, the difference of fourth-order Taylor polynomials evaluates to

$$\mathbb{E}[P_4(\epsilon Y)] - \sum_{i=1}^{d} \mathbb{E}[P_4(\epsilon U_i)] = \frac{\epsilon^4}{12}\Big(\mathbb{E}[Y^4] - \sum_i \mathbb{E}[U_i^4]\Big) = \frac{\epsilon^4}{2}\binom{d}{2}(e-1)^2 = \frac{d(d-1)(e-1)^2}{4(d+1)^4}. \tag{19}$$

Expanding for large d,

$$\frac{d(d-1)}{(d+1)^4} = \frac{1}{d^2} + O(d^{-3}),$$

so the leading term is $(e-1)^2/(4d^2) + O(d^{-3})$.

Taylor remainder. On A_d,

$$\mathbb{E}\big[|R_4(\epsilon Y)|\mathbf{1}_{A_d}\big] \leqslant \tfrac{4}{5}\epsilon^5\mathbb{E}[|Y|^5] = O(d^{-5}) \cdot O(d^{5/2}) = O(d^{-5/2}).$$

The marginal remainders $|R_4(\epsilon U_i)| \leqslant \frac{4}{5}\epsilon^5|U_i|^5$ each have expectation $O(\epsilon^5) = O(d^{-5})$, and summing over i gives $O(d^{-4})$. Total Taylor remainder is $O(d^{-5/2})$.

Removing the typical-set indicator. We must replace $\mathbb{E}[P_4(\epsilon Y)\mathbf{1}_{A_d}]$ with $\mathbb{E}[P_4(\epsilon Y)]$. The dominant degree-4 term satisfies, by Cauchy–Schwarz and Rosenthal,

$$\mathbb{E}[Y^4\mathbf{1}_{A_d^c}] \leqslant \sqrt{\mathbb{E}[Y^8] \cdot \mathbb{P}(A_d^c)} = \sqrt{O(d^4) \cdot O(d^{-5})} = O(d^{-1/2}).$$

Multiplying by $\epsilon^4 = O(d^{-4})$ gives $O(d^{-9/2})$. Lower-degree terms contribute even less. The same bound applies to the marginal sum. Total contribution from removing the indicator: $O(d^{-9/2})$.

C.7 Putting It Together

Combining Eqs. (14), (16), (17), (19), and the remainder bounds,

$$\begin{aligned}
\mathrm{TC}(S) &= \mathbb{E}_Z[g(\epsilon Y)] - \sum_{i=1}^{d} \mathbb{E}_Z[g(\epsilon U_i)] \\
&= \Big(\mathbb{E}_Z[P_4(\epsilon Y)] - \sum_{i=1}^{d} \mathbb{E}_Z[P_4(\epsilon U_i)]\Big) + O(d^{-5/2}) + O(d^{-5/2}) + O(d^{-9/2}) \\
&= \frac{(e-1)^2}{4d^2} + O(d^{-5/2}),
\end{aligned}$$

which completes the proof. □

D Proof of Theorem 4.1

We restate the theorem for convenience. Let $Z \sim \mathcal{N}(0, I_d)$ and $\Delta \sim \mathrm{Uniform}\{\mathbf{0}, \mathbf{u}_1, \ldots, \mathbf{u}_d\}$ be independent, and set $S = Z + \Delta$. Then

$$D_{\mathrm{KL}}(S \,\|\, Z) = \frac{e-1}{2d} + O(d^{-3/2}) \qquad (d \to \infty).$$

D.1 Setup

We reuse the notation from Appendix C. Let ϕ denote the standard d-dimensional Gaussian density. The likelihood ratio

$$L(z) = \frac{P_S(z)}{\phi(z)} = 1 + \epsilon Y, \qquad \text{with } \epsilon = \frac{1}{d+1}, \quad Y = \sum_{i=1}^{d} U_i, \quad U_i = e^{Z_i - 1/2} - 1,$$

satisfies $\mathbb{E}[U_i] = 0$, $\mathbb{E}[U_i^2] = e - 1$, and Rosenthal's inequality gives $\mathbb{E}[|Y|^p] = O(d^{p/2})$ for any $p \geqslant 2$.

By the change-of-measure identity applied to the joint KL,

$$D_{\mathrm{KL}}(S \,\|\, Z) = \mathbb{E}_S[\ln L(S)] = \mathbb{E}_Z[L(Z) \ln L(Z)] = \mathbb{E}_Z[g(\epsilon Y)], \tag{20}$$

where $g(u) = (1+u)\ln(1+u)$. This is precisely the first term in the total-correlation identity (14).

D.2 Typical Set and Tail Bound

Define $A_d = \{|\epsilon Y| < 1/2\}$. Markov's inequality with $p = 6$ gives

$$\mathbb{P}(A_d^c) \leqslant 2^6\epsilon^6\mathbb{E}[|Y|^6] = O(d^{-3}).$$

Using the bound $g(y)^2 \leqslant 2(y^2 + y^4) + 2$ for all $y > -1$ and the moment bounds, $\mathbb{E}[g(\epsilon Y)^2] = O(1)$. Cauchy–Schwarz gives

$$\big|\mathbb{E}[g(\epsilon Y)\mathbf{1}_{A_d^c}]\big| \leqslant \sqrt{\mathbb{E}[g(\epsilon Y)^2]} \cdot \sqrt{\mathbb{P}(A_d^c)} = O(d^{-3/2}). \tag{21}$$

D.3 Taylor Expansion to Second Order

On A_d, $|\epsilon Y| < 1/2$, so we may Taylor-expand g to second order. Using $g(0) = 0$, $g'(0) = 1$, $g''(0) = 1$, and $|g'''(\xi)| \leqslant |1/(1+\xi)^2| \leqslant 4$ for $|\xi| \leqslant 1/2$, Lagrange's remainder gives

$$g(u) = u + \tfrac{1}{2}u^2 + R_2(u), \qquad |R_2(u)| \leqslant \tfrac{4}{6}|u|^3 = \tfrac{2}{3}|u|^3 \tag{22}$$

for $|u| \leqslant 1/2$.

First-order term. $\mathbb{E}[\epsilon Y] = \epsilon \sum_i \mathbb{E}[U_i] = 0$, so the first-order term contributes nothing.

Second-order term. Using $\mathbb{E}[Y^2] = d(e-1)$ by independence and $\mathbb{E}[U_i^2] = e - 1$,

$$\frac{1}{2}\mathbb{E}[(\epsilon Y)^2] = \frac{\epsilon^2}{2} \cdot d(e-1) = \frac{d(e-1)}{2(d+1)^2}.$$

For large d, $d/(d+1)^2 = 1/d \cdot (d/(d+1))^2 = 1/d \cdot (1 - 1/(d+1))^2 = 1/d - 2/d^2 + O(d^{-3})$, so

$$\frac{1}{2}\mathbb{E}[(\epsilon Y)^2] = \frac{e-1}{2d} + O(d^{-2}). \tag{23}$$

Taylor remainder. On A_d,

$$\mathbb{E}\big[|R_2(\epsilon Y)|\mathbf{1}_{A_d}\big] \leqslant \tfrac{2}{3}\epsilon^3 \mathbb{E}[|Y|^3] = O(d^{-3}) \cdot O(d^{3/2}) = O(d^{-3/2}).$$

Removing the typical-set indicator. The dominant degree-2 term satisfies, by Cauchy–Schwarz and Rosenthal,

$$\mathbb{E}[Y^2 \mathbf{1}_{A_d^c}] \leqslant \sqrt{\mathbb{E}[Y^4] \cdot \mathbb{P}(A_d^c)} = \sqrt{O(d^2) \cdot O(d^{-3})} = O(d^{-1/2}).$$

Multiplying by $\epsilon^2 = O(d^{-2})$ gives $O(d^{-5/2})$. The first-order term contributes similarly. Total cost of removing the indicator is $O(d^{-5/2})$, smaller than the Taylor remainder.

D.4 Putting It Together

Combining Eqs. (20), (21), (23), and the remainder bounds,

$$\begin{aligned} D_{\mathrm{KL}}(S \,\|\, Z) &= \mathbb{E}_Z[g(\epsilon Y)] \\ &= \mathbb{E}_Z\big[(P_2(\epsilon Y) + R_2(\epsilon Y))\mathbf{1}_{A_d}\big] + \mathbb{E}_Z[g(\epsilon Y)\mathbf{1}_{A_d^c}] \\ &= \mathbb{E}_Z[P_2(\epsilon Y)] + O(d^{-3/2}) + O(d^{-5/2}) + O(d^{-3/2}) \\ &= \frac{e-1}{2d} + O(d^{-3/2}), \end{aligned}$$

where $P_2(u) = u + u^2/2$ and the dominant error is the Taylor remainder. □

E Extending total correlation decay to general Δ, Z

Let $Z \sim \mathcal{N}(0, \mathrm{diag}(\sigma_1^2, \ldots, \sigma_d^2))$ and $\Delta \sim \mathrm{Uniform}\{\mathbf{0}, \delta_1 \mathbf{u}_1, \ldots, \delta_d \mathbf{u}_d\}$ be independent. Define the standardized intervention strengths for each coordinate:

$$\tau_i = \frac{\delta_i}{\sigma_i}, \qquad i = 1, \ldots, d,$$

and assume $\max_i |\tau_i| \leqslant M$ uniformly in d, with $\tau_i \neq 0$. Set $\epsilon = 1/(d+1)$ and

$$U_i = e^{\tau_i \frac{Z_i}{\sigma_i} - \frac{\tau_i^2}{2}} - 1.$$

Because $Z_i/\sigma_i \sim \mathcal{N}(0,1)$, each U_i is a centered log-normal variable; its moments are

$$\mathbb{E}[U_i] = 0, \qquad a_i := \mathbb{E}[U_i^2] = e^{\tau_i^2} - 1,$$

and all moments $\mathbb{E}[|U_i|^p]$ are finite, bounded by constants depending only on M and p.

Let $Y = \sum_{i=1}^d U_i$. Exactly as in the previous case, the joint and marginal likelihood ratios become

$$L(Z) = 1 + \epsilon Y, \qquad L_i(Z_i) = 1 + \epsilon U_i,$$

and the exact identity (14) carries over with no change:

$$\mathrm{TC}(S) = \mathbb{E}_Z[g(\epsilon Y)] - \sum_{i=1}^d \mathbb{E}_Z[g(\epsilon U_i)], \qquad g(u) = (1+u)\ln(1+u).$$

Rosenthal's inequality applied to the independent (but not identically distributed) variables U_i still gives

$$\mathbb{E}[|Y|^p] \leqslant C'_p\, d^{p/2}, \qquad p \geqslant 2,$$

because $\sum_i \mathbb{E}[U_i^2] \asymp d$ and higher moments are uniformly bounded. Defining the typical set $A_d = \{|\epsilon Y| < \frac{1}{2}\}$ and choosing $p = 10$, we obtain the same tail estimate

$$\mathbb{P}(A_d^c) = O(d^{-5}), \qquad \left|\mathbb{E}[g(\epsilon Y)\mathbf{1}_{A_d^c}]\right| = O(d^{-5/2}),$$

and the marginal remainders are similarly negligible ($O(d^{-4})$ after summation).

On A_d we expand g to fourth order, i.e., $g(u) = u + \frac{1}{2}u^2 - \frac{1}{6}u^3 + \frac{1}{12}u^4 + O(u^5)$. The first three moments again cancel perfectly because the U_i are independent and centered:

$$\mathbb{E}[\epsilon Y] = 0, \quad \mathbb{E}[(\epsilon Y)^2] = \epsilon^2 \sum_i a_i = \sum_i \mathbb{E}[(\epsilon U_i)^2], \quad \mathbb{E}[(\epsilon Y)^3] = \sum_i \mathbb{E}[(\epsilon U_i)^3].$$

Only the fourth order terms survive, and as before the expectation over the atypical set contributes $O(d^{-9/2})$. Thus

$$\mathrm{TC}(S) = \frac{\epsilon^4}{12}\Big(\mathbb{E}[Y^4] - \sum_i \mathbb{E}[U_i^4]\Big) + O(d^{-5/2}).$$

Since $\mathbb{E}[Y^4] = \sum_i \mathbb{E}[U_i^4] + 6\sum_{i<j} a_i a_j$, we obtain

$$\mathrm{TC}(S) = \frac{\epsilon^4}{2} \sum_{i<j} a_i a_j + O(d^{-5/2}).$$

Expanding $\epsilon^4 = d^{-4} - 4d^{-5} + O(d^{-6})$ and noting $\sum_{i<j} a_i a_j = \frac{1}{2}\big((\sum_i a_i)^2 - \sum_i a_i^2\big)$, with $\sum_i a_i^2 = O(d)$, the leading term simplifies to

$$\mathrm{TC}(S) = \frac{1}{4d^4}\Big(\sum_{i=1}^d a_i\Big)^2 + O(d^{-3}),$$

and the Taylor remainder $O(d^{-5/2})$ dominates the $O(d^{-3})$ error. Using the empirical mean $\bar{a} = \frac{1}{d}\sum_i a_i$ we finally have

$$\mathrm{TC}(S) = \frac{\bar{a}^2}{4d^2} + O(d^{-5/2}), \qquad \bar{a} = \frac{1}{d}\sum_{i=1}^d \big(e^{\tau_i^2} - 1\big).$$

This is a generalization of the simplified result when all $\sigma_i = \delta_i = 1$ so that $\tau_i = 1$.

The same steps applied only to the joint term yield

$$D_{\mathrm{KL}}(S \,\|\, Z) = \mathbb{E}_Z[g(\epsilon Y)] = \frac{\bar{a}}{2d} + O(d^{-3/2}),$$

F Kurtosis–Recovery Ablation Under the Single-Node Design

Section 4.2 demonstrates that ICA-LiNGAM's recovery quality tracks per-coordinate non-Gaussianity under *independent-occurrence* interventions, where the pooled noise coordinates are mutually independent by construction and only the non-Gaussianity violation is active. A natural follow-up question is whether the same kurtosis–recovery relationship persists in the realistic *single-node* design where both violations—dependence and reduced non-Gaussianity—are present simultaneously. This appendix confirms that it does.

Setup. We use the synthetic configuration of Section 5.1: 100 random Erdős–Rényi DAGs with $d = 10$ nodes, expected edge count 10, and edge weights drawn from $[-2, -0.5] \cup [0.5, 2]$. Data is pooled across d single-node intervention environments. We sweep the per-source minimum excess kurtosis by varying the intervention shift magnitude, and report mean structural Hamming distance against the true DAG for ICA-LiNGAM and the label-aware baselines (GIES, IGSP, UT-IGSP) used in the main paper.

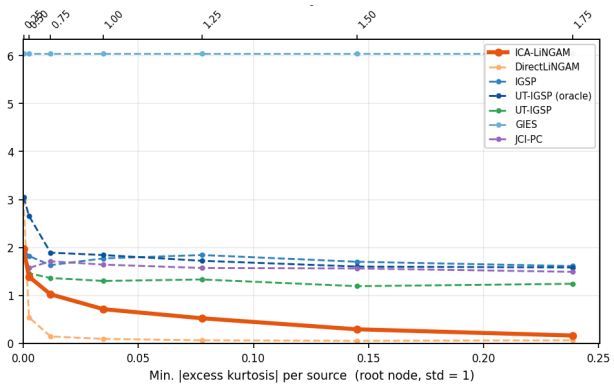


Figure 5: Mean SHD versus minimum per-source excess kurtosis on $d = 10$ ER DAGs under the single-node intervention design. ICA-LiNGAM's SHD falls sharply as kurtosis grows; label-aware baselines (GIES, IGSP, UT-IGSP) are largely insensitive because they rely on conditional-independence tests across environments rather than non-Gaussianity. This confirms that the sweet-spot phenomenon characterized in Section 4.2 persists when dependence is also present.

Results. Figure 5 shows that ICA-LiNGAM's SHD falls sharply with increasing kurtosis even under the single-node design, mirroring the trend established in Section 4.2 for independent-occurrence interventions. The label-aware baselines remain largely insensitive to kurtosis: their recovery quality is bounded by the per-environment sample size and the discriminative power of their invariance tests, neither of which depends on the non-Gaussianity of the noise. This contrast illustrates that ICA-LiNGAM and the label-aware methods exploit fundamentally different signals—higher-order moments versus environment-conditional independence—which is consistent with the complementary scaling regimes characterized in Sections 3.1 and 4.1.

G Comparison With Labels-Only Methods

The baselines in Section 5 mostly receive both environment labels and intervention targets. This appendix places ICA-LiNGAM within the full ladder of side information: no labels and no targets (ICA-LiNGAM); environment labels but unknown targets (JCI with environment indicators, UT-IGSP, Ψ-FCI); and labels with known targets (JCI with target indicators, IGSP, GIES).

Methods. *JCI (labels only).* Following Mooij et al. [2020], utilize additional variables that represent which environment each datapoint comes from. That is, we append one binary "context indicator" per non-reference environment and run PC with Fisher-z tests at $\alpha = 0.05$ on the augmented data. The JCI background knowledge is imposed throughout: context variables are exogenous (no edges from system→context), all context variables are mutually adjacent, and every conditional-independence test between system variables also conditions on the context variables. The method is told which samples share an environment, but not which nodes each environment targets. *JCI (known targets)* instead uses one indicator per intervened node, connected only to that node.

Ψ-FCI. We use the `dodiscover` implementation [PyWhy Community, 2022] with one F-node per non-reference environment. Within-environment conditional-independence tests use Fisher-z, and the cross-environment conditional-discrepancy tests use a linear test matching the mean-shift model: a partial-correlation test of the environment indicator against the target given the conditioning set ($\alpha = 0.05$). We do not use the default kernel discrepancy test, which is computationally prohibitive at Sachs sample sizes. *UT-IGSP, IGSP, GIES* are run as in Section 5.

SHD conventions. Figures 3 and 4 report SHD as computed by `gcastle` [Zhang et al., 2021], which does not penalize an edge left undirected when the skeleton is correct. For the PAG returned by Ψ-FCI, we map every edge without a definite orientation (bidirected or with a circle endpoint) to an undirected edge before applying this convention. Because this favors methods that leave edges unoriented, we also report a *strict* SHD, in which each missing edge, extra edge, reversed edge, and edge without a definite correct orientation costs 1, and the F1 score over directed edges. ICA-LiNGAM always returns a fully directed DAG, so its two SHDs coincide.

G.1 Synthetic Data Under the Single-Node Design

Setup. We use 100 Erdős–Rényi DAGs with $d = 10$ nodes, 10 edges, and edge weights with $|w| \in [0.5, 2]$. Data come from the single-node design: one environment per node, each shifting the mean of its target by $\delta \widehat{\mathrm{Var}}(X_i)$ with $\delta = 1$. We add 100 observational samples, which serve as JCI's reference environment. Interventional sample sizes $n \in \{200, 500, 1000, 2000, 5000, 10000\}$ are nested subsamples of one random permutation of the pool. We also ran JCI at $\alpha = 0.01$; the results differ from $\alpha = 0.05$ by at most 0.2 SHD.

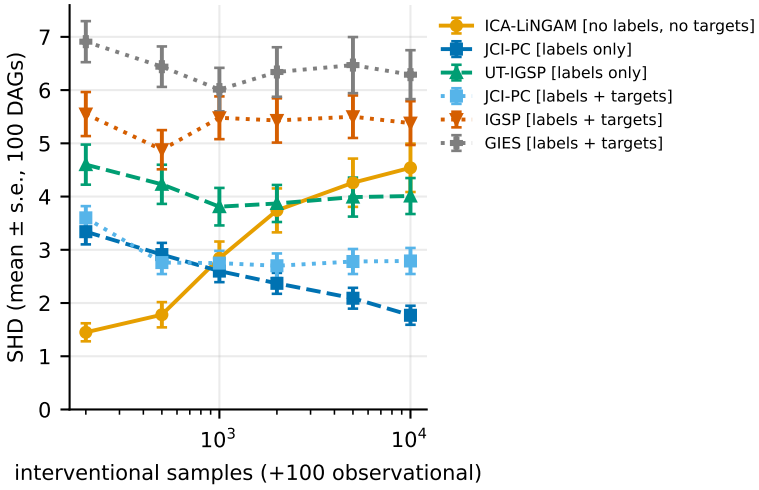


Figure 6: SHD (`gcastle` convention) under the single-node design, $d = 10$, mean ± s.e. over 100 DAGs. Line style encodes side information: solid, none; dashed, environment labels only; dotted, labels and intervention targets. Under strict SHD all curves are unchanged except labels-only JCI, which rises by at most 0.25 (at $n = 200$).

Results. Figure 6 shows two regimes. At small n, ICA-LiNGAM has the lowest SHD of all methods (1.45 at $n = 200$, versus 3.34 for labels-only JCI and 3.60 for JCI with known targets). As n grows,

ICA-LiNGAM's SHD rises to 4.54 at $n = 10^4$, while labels-only JCI falls to 1.77; the curves cross near $n = 10^3$. The degradation comes from extra edges rather than from the causal order: in a check on 30 of these DAGs, the order estimated by ICA-LiNGAM was fully correct more often at $n = 10^4$ than at $n = 200$ (70% versus 57%), yet even among DAGs with a correct order the number of extra edges rose from 0.35 to 4.33. ICA-LiNGAM prunes edges by regressing each variable on its predecessors in the estimated order, which assumes independent noise; under the atomic design the noise terms are correlated (Section 2.2), so non-parents receive small but nonzero population coefficients, which survive pruning once n is large enough to estimate them precisely. Because the 100 observational samples make up a shrinking fraction of the pool as n grows, we also reran ICA-LiNGAM with the observational share held at $1/11$ (that is, $n/10$ observational samples, the size of an average interventional environment). Its SHD still rises with n, from 1.84 at $n = 200$ to 4.21 at $n = 10^4$, so the degradation is not an artifact of the changing observational fraction. Labels-only JCI outperforms JCI with known targets at large n (1.77 versus 2.79). We have not investigated why; the two implementations differ in how contexts are encoded and in their reference environment. IGSP and GIES plateau between 4.9 and 6.9, above ICA-LiNGAM at every sample size. UT-IGSP plateaus near 4, below ICA-LiNGAM for $n \geqslant 5000$.

G.2 Sachs Protein Signaling Data

Setup. We use the same preprocessing, ground-truth network (17 edges), and sampling scheme as Figure 4: at each total sample size n we draw n samples uniformly without replacement from the pool, 30 times, and use the full pool as the last point. JCI and Ψ-FCI are deterministic, so on the full pool we report a single run. We run both methods on the 9-condition pool used by ICA-LiNGAM (6456 samples) and on the 5-condition subset used by GIES and IGSP (4275 samples), with the CD3/CD28 condition as the reference environment.

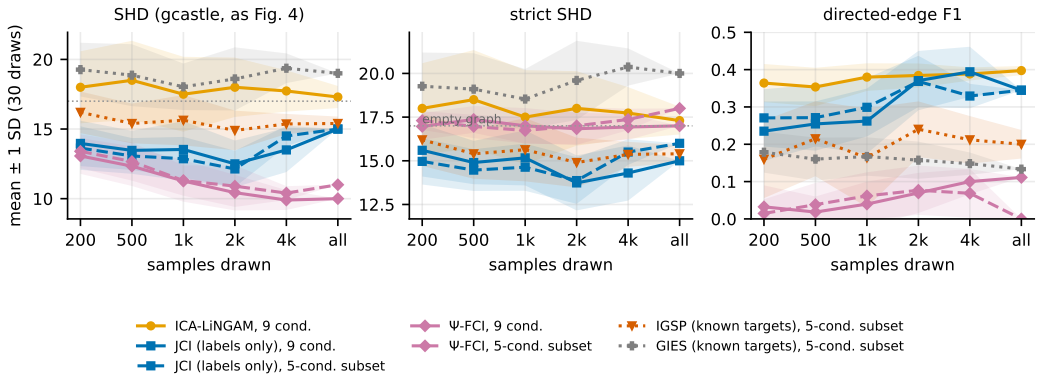


Figure 7: ICA-LiNGAM (no labels) versus labels-only JCI and Ψ-FCI and known-target GIES and IGSP on Sachs, mean ± 1 SD over 30 draws. Solid lines: 9-condition pool; dashed: labels-only methods on the 5-condition subset; dotted: known-target methods on the 5-condition subset. The horizontal dotted line marks the SHD of the empty graph (17).

Results. Under the `gcastle` convention, both labels-only methods have lower SHD than ICA-LiNGAM at every sample size (Figure 7, left). The gap reflects a better skeleton more than better orientation. For $n \geqslant 1000$, Ψ-FCI returns on average 4–8 bidirected edges and fewer than three directed edges. Its strict SHD stays between 16.7 and 18, around the SHD of the empty graph (17), and its directed-edge F1 never exceeds 0.11. JCI also improves on ICA-LiNGAM in strict SHD (13.7–16 versus 17.3–18.5). ICA-LiNGAM has the highest directed-edge F1 up to $n = 1000$; at $n = 2000$ and $n = 4000$, JCI matches it (0.37–0.39 versus 0.38–0.39 on the 9-condition pool). Of the known-target methods, only IGSP has lower SHD than ICA-LiNGAM (15.6 versus 17.5 at $n = 1000$), and both orient fewer edges correctly: directed-edge F1 is at most 0.24 for IGSP and 0.18 for GIES, against 0.35–0.40 for ICA-LiNGAM. Knowing the intervention targets therefore does not, on Sachs, translate into better orientation than pooling without labels. The prevalence of bidirected edges in Ψ-FCI's output is consistent with latent confounding among the measured proteins, which would violate the causal-sufficiency assumption made by ICA-LiNGAM, JCI, GIES, and IGSP alike.

Table 1: Sachs results at $n = 1000$ and on the full pool (mean $\pm$ 1 SD over 30 draws; single run where no SD is shown). Dir. F1 is the F1 score over directed edges. GIES and IGSP receive known targets; IGSP is randomized, so it is run on every draw even on the full pool.

	$n = 1000$			full pool		
Method, pool	SHD	strict SHD	Dir. F1	SHD	strict SHD	Dir. F1
ICA-LiNGAM, 9 cond. (no labels)	17.5 ± 2.7	17.5 ± 2.7	0.38	17.3 ± 0.8	17.3 ± 0.8	0.40
JCI (labels only), 9 cond.	13.5 ± 2.0	15.2 ± 1.6	0.26	15	15	0.34
JCI (labels only), 5-cond. subset	12.9 ± 1.6	14.6 ± 1.4	0.30	15	16	0.34
Ψ-FCI, 9 cond.	11.3 ± 1.5	17.0 ± 1.0	0.04	10	17	0.11
Ψ-FCI, 5-cond. subset	11.3 ± 1.1	16.7 ± 0.7	0.06	11	18	0.00
IGSP (known targets), 5-cond. subset	15.6 ± 1.8	15.6 ± 1.8	0.16	15.4 ± 0.5	15.4 ± 0.5	0.20
GIES (known targets), 5-cond. subset	18.0 ± 1.7	18.5 ± 1.7	0.17	19	20	0.13

G.3 Noisy Labels

Setup. We use the setting of Appendix G.1 at $n = 10^3$ and $n = 10^4$ and corrupt the side information given to the label-aware methods in two ways. Under *environment-label noise*, each interventional sample is assigned to a uniformly chosen wrong interventional environment with probability ε; observational samples keep their label. This corrupts the grouping that every label-aware method relies on. Under *target noise*, the grouping is correct, but each environment's listed target is replaced by a uniformly chosen wrong node with probability ε, which affects only the methods that use targets. Corruption is nested in ε: a sample or environment corrupted at one rate is also corrupted at every higher rate. Mislabeling leaves the data unchanged, so ICA-LiNGAM, which uses no labels, serves as a fixed reference. At $\varepsilon = 0$ the results reproduce Figure 6.

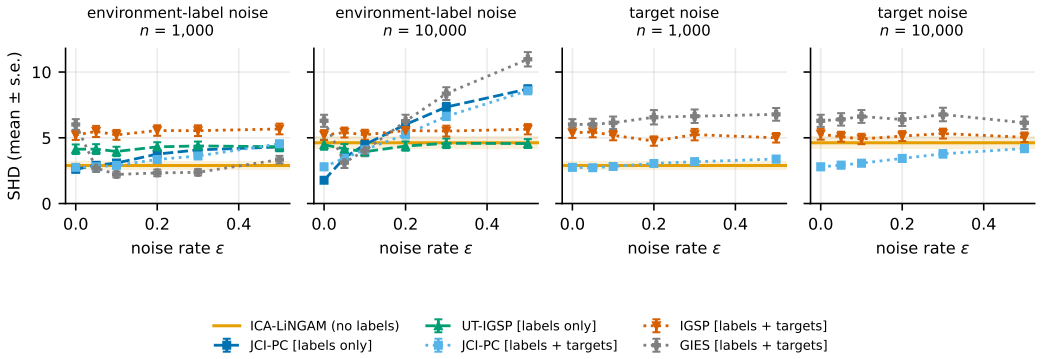


Figure 8: SHD (`gcastle` convention) under the single-node design as the side information given to label-aware methods is corrupted, mean $\pm$ s.e. over 100 DAGs. The solid horizontal line is ICA-LiNGAM, which uses no labels. Line style encodes side information as in Figure 6.

Results. Environment-label noise quickly removes the advantage of the labels-only methods (Figure 8, left two panels). At $n = 10^4$, where clean labels give JCI an SHD of 1.77 against ICA-LiNGAM's 4.62, JCI loses its advantage once 10–20% of samples are mislabeled, and at $\varepsilon = 0.5$ its SHD reaches 8.71. At $n = 10^3$ the crossover occurs between $\varepsilon = 0.05$ and $\varepsilon = 0.1$. JCI with known targets behaves similarly. Target noise is much less harmful (right two panels): JCI with known targets degrades gradually, crossing ICA-LiNGAM between $\varepsilon = 0.1$ and $\varepsilon = 0.2$ at $n = 10^3$ and remaining below it up to $\varepsilon = 0.5$ at $n = 10^4$ (4.18 versus 4.62). IGSP and UT-IGSP change by less than 0.7 under either kind of noise. GIES behaves non-monotonically: moderate environment-label noise lowers its SHD (from 6.01 to 2.21 at $n = 10^3$, $\varepsilon = 0.1$), while heavy noise raises it to 10.97 at $n = 10^4$. GIES assumes hard interventions whereas ours are soft mean shifts, which may explain this; we have not investigated it further.

In this setting, the large-sample advantage of label-aware methods therefore depends on the environment labels being nearly exact; errors in the listed targets matter much less than errors in which

samples are grouped together. This experiment isolates label noise, leaving the data unchanged. Errors that also change the data, such as correlated off-target effects, degrade ICA-LiNGAM as well (Appendix H).

H Robustness Ablations

This appendix reports ablations on how sensitive ICA-LiNGAM is to departures from our modeling assumptions. All experiments use the synthetic configuration of Section 5.1: 100 random Erdős–Rényi DAGs with $d = 10$ nodes and expected edge count 10. We report means $\pm$ standard errors over the 100 DAGs. Where shown, DirectLiNGAM [Shimizu et al., 2011] is included as a second LiNGAM estimator.

H.1 Dependence Versus Recovery Under the Single-Node Design

To relate source dependence to graph recovery, we run ICA-LiNGAM under single-node interventions, so that total correlation is necessarily nonzero. Each DAG uses 2,000 interventional and 100 observational samples. We vary the intervention strength δ, which sets the mean shift on target i to $\delta \widehat{\mathrm{Var}}(X_i)$, where $\widehat{\mathrm{Var}}(X_i)$ is the empirical observational variance of X_i, and the target-probability imbalance $r = \max_k \pi_k / \min_k \pi_k \in \{1, 2, 5, 10\}$, where π_k is the probability of targeting node k. Total correlation is computed from the true pooled disturbances (oracle).

Table 2: ICA-LiNGAM under single-node interventions with varying intervention strength δ and target-probability imbalance r.

δ	r	SHD	Oracle total correlation	Mean excess kurtosis
1	1	3.58 ± 0.40	0.131 ± 0.008	1.30 ± 0.06
1	2	3.65 ± 0.39	0.134 ± 0.008	1.36 ± 0.07
1	5	3.65 ± 0.40	0.139 ± 0.009	1.65 ± 0.10
1	10	3.68 ± 0.39	0.144 ± 0.010	2.11 ± 0.16
2	1	7.33 ± 0.56	0.350 ± 0.010	2.17 ± 0.06
2	2	7.28 ± 0.57	0.354 ± 0.011	2.27 ± 0.07
2	5	6.98 ± 0.50	0.361 ± 0.011	2.83 ± 0.12
2	10	7.15 ± 0.47	0.367 ± 0.012	3.67 ± 0.19
5	1	17.73 ± 0.48	0.891 ± 0.003	3.71 ± 0.03
5	2	17.42 ± 0.45	0.889 ± 0.003	3.95 ± 0.05
5	5	15.71 ± 0.45	0.880 ± 0.003	5.09 ± 0.10
5	10	13.98 ± 0.44	0.869 ± 0.003	6.78 ± 0.18
10	1	21.56 ± 0.36	0.948 ± 0.000	4.62 ± 0.01
10	2	20.63 ± 0.38	0.945 ± 0.000	5.01 ± 0.02
10	5	19.29 ± 0.46	0.934 ± 0.000	6.81 ± 0.05
10	10	18.27 ± 0.46	0.920 ± 0.000	9.53 ± 0.11

At $\delta = 1$, increasing the imbalance changes total correlation only modestly (from 0.131 to 0.144) and SHD stays essentially constant, so small changes in dependence are tolerated. Across intervention strengths, however, total correlation and SHD rise sharply together: with balanced targeting, increasing δ from 1 to 10 raises total correlation from 0.131 to 0.948 and SHD from 3.58 to 21.56. Dependence becomes harmful when the mutually exclusive shifts are strong enough to substantially violate the independent-source model, even though excess kurtosis also increases.

H.2 Sensitivity to Other Assumption Violations

Ablations A–C vary one factor while keeping the sources independent (independent-occurrence interventions); Ablation D instead directly induces dependence through correlated off-target effects. Each uses 2,000 interventional samples per DAG, with all non-varied parameters held fixed. All shifts are scaled by the observational variance of the target, as in Section 5.1. The $\tau > 0$ rows of Ablation B come from a separate run; rerunning the shared baseline in that run gives ICA-LiNGAM an SHD of 1.22 ± 0.17, within run-to-run variation of the 1.30 reported below (DirectLiNGAM is

deterministic and gives 0.60 in both runs). Because A–C keep the sources independent, their baseline SHD for ICA-LiNGAM (1.30) is lower than the single-node baseline in Table 2 (3.58).

(A) Unequal environment sizes. Per-node intervention probabilities are varied while their mean is held at 0.5, so the expected number of intervention events is fixed. Node i is intervened on with probability $p_i = 0.5 + h\, z_{\sigma(i)}$, where $z_1, \ldots, z_d$ are d evenly spaced values in $[-1, 1]$ and σ is a random permutation drawn once per DAG. The probabilities therefore range over $[0.5 - h, 0.5 + h]$ with mean 0.5; at $h = 0.45$, the most frequently targeted node is intervened on 19 times as often as the least.

(B) Heterogeneous intervention strengths. Node-specific shift magnitudes are varied while the RMS shift energy is held fixed. Node i receives the shift $\delta_i\, \delta\, \widehat{\mathrm{Var}}(X_i)$ with $\delta_i = e^{\tau z_{\sigma(i)}} / \left(\frac{1}{d}\sum_j e^{2\tau z_{\sigma(j)}}\right)^{1/2}$, with z and σ as in (A), so that $\frac{1}{d}\sum_i \delta_i^2 = 1$ at every τ. The strongest-to-weakest multiplier ratio is $e^{2\tau}$, which grows from 1 to approximately 7.4.

(C) Uniformly weak interventions. The multiplier δ in the mean shift $\delta\, \widehat{\mathrm{Var}}(X_i)$ is reduced from 1 to 0.1.

(D) Correlated off-target effects. Each intended intervention also shifts a fixed off-target node through the same Bernoulli indicator, directly coupling the effective disturbances; ρ controls the strength of this coupling. Let π be a random derangement of the nodes ($\pi(i) \neq i$), drawn once per DAG, and let B_i be node i's intervention indicator. An intervention on i shifts node i by $c(\rho)\, \Delta_i B_i$ and node $\pi(i)$ by $c(\rho)\, \rho\, \Delta_i B_i$, where $\Delta_i = \delta\, \widehat{\mathrm{Var}}(X_i)$ is the baseline shift and $c(\rho) = (1 + \rho^2)^{-1/2}$ keeps the total shift energy of each intervention fixed.

Table 3: Robustness ablations. Means ± standard errors over 100 DAGs with $d = 10$.

		DirectLiNGAM		ICA-LiNGAM	
Ablation	Severity	SHD	F1	SHD	F1
(A) Unequal environment sizes	$h = 0.00$	0.60 ± 0.10	0.964 ± 0.006	1.30 ± 0.17	0.910 ± 0.011
	$h = 0.15$	0.64 ± 0.11	0.967 ± 0.005	1.37 ± 0.18	0.914 ± 0.010
	$h = 0.30$	0.81 ± 0.12	0.959 ± 0.006	1.78 ± 0.23	0.895 ± 0.011
	$h = 0.45$	1.21 ± 0.17	0.944 ± 0.007	2.15 ± 0.23	0.876 ± 0.012
(B) Heterogeneous strengths	$\tau = 0.00$	0.60 ± 0.10	0.964 ± 0.006	1.30 ± 0.17	0.910 ± 0.011
	$\tau = 0.35$	0.66 ± 0.11	0.961 ± 0.006	1.16 ± 0.14	0.924 ± 0.009
	$\tau = 0.70$	0.83 ± 0.12	0.945 ± 0.008	1.40 ± 0.18	0.911 ± 0.010
	$\tau = 1.00$	1.05 ± 0.13	0.919 ± 0.010	1.46 ± 0.17	0.901 ± 0.011
(C) Uniformly weak shifts	$\delta = 1.00$	0.60 ± 0.10	0.964 ± 0.006	1.30 ± 0.17	0.910 ± 0.011
	$\delta = 0.75$	0.66 ± 0.10	0.948 ± 0.008	1.10 ± 0.15	0.919 ± 0.010
	$\delta = 0.50$	1.17 ± 0.17	0.897 ± 0.015	1.42 ± 0.19	0.892 ± 0.014
	$\delta = 0.25$	4.41 ± 0.29	0.666 ± 0.020	1.77 ± 0.21	0.867 ± 0.013
	$\delta = 0.10$	6.58 ± 0.40	0.550 ± 0.019	2.42 ± 0.25	0.832 ± 0.014
(D) Correlated off-target	$\rho = 0.00$	0.60 ± 0.10	0.964 ± 0.006	1.30 ± 0.17	0.910 ± 0.011
	$\rho = 0.10$	2.31 ± 0.30	0.883 ± 0.014	2.86 ± 0.24	0.853 ± 0.011
	$\rho = 0.25$	4.80 ± 0.31	0.767 ± 0.013	5.11 ± 0.33	0.749 ± 0.013
	$\rho = 0.50$	8.27 ± 0.33	0.629 ± 0.011	8.42 ± 0.42	0.631 ± 0.013
	$\rho = 1.00$	10.08 ± 0.40	0.556 ± 0.015	10.70 ± 0.50	0.550 ± 0.014

Results. Unequal environment sizes (A) degrade both methods gradually, even when the most frequently targeted node is intervened on 19 times as often as the least. Heterogeneous strengths (B) degrade both methods only modestly: at $\tau = 1$, where the strongest shift is 7.4 times the weakest, SHD rises from 0.60 to 1.05 for DirectLiNGAM and from 1.30 to 1.46 for ICA-LiNGAM. Under uniformly weak shifts (C), ICA-LiNGAM degrades more gracefully than DirectLiNGAM. Correlated off-target effects (D) are the most damaging violation: both methods exceed an SHD of 8 at $\rho = 0.5$ and approach 10 at $\rho = 1$. This is why we describe the atomic design as an adverse reference case among designs without excess co-occurrence, rather than as a worst case over all designs: shared intervention leakage can induce a more severe violation of source independence.

I Computational Resources

All experiments can be conducted on an M2 MacBook Air using CPU. The synthetic data experiment on 100 Erdős–Rényi DAGs was run on a server using 64 CPU cores and took about 48 hours to run. The rest of the experiments were run on CPU and each took $12 - 72$ hours. The camera-ready experiments (Appendix G and the rerun of Ablation B in Appendix H) ran on a 10-core laptop CPU; the longest, Ψ-FCI on Sachs, took about two hours.

J Code Availability

The code to reproduce the experiments is available at https://github.com/sophiaxxiao/NEURIPS-2026-Causal-Discovery-from-Unknown-Interventions.

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